

Lesson Notes: Conversations in Scratch Using Broadcasts

1. Lesson Introduction

In Scratch, sprites can **talk to each other** and work together. This is called a **conversation**. Sprites communicate by sending messages called **broadcasts**. Broadcasts help sprites react at the right time, making stories, animations, and games more interactive.

2. Learning Objectives

By the end of the lesson, learners should be able to:

1. Explain what a **broadcast** is in Scratch.
2. Create a simple **conversation** between two sprites.
3. Use **broadcast** and **when I receive** blocks correctly.
4. Control the order of speech using broadcasts.
5. Design a short animated dialogue using Scratch.

3. Key Vocabulary

- **Broadcast** – a message sent to all sprites in a project.
- **When I receive** – a block that makes a sprite react to a broadcast.
- **Conversation** – sprites talking to each other in turns.
- **Sprite** – a character in Scratch.

4. Explanation: What is a Broadcast?

A **broadcast** is a message that one sprite sends and other sprites receive. It is found under the **Events** blocks.

- One sprite **broadcasts** a message.
- Other sprites respond using **when I receive [message]**.
- Broadcasts help control **timing and order**.

☐ Think of a broadcast like **calling someone's name** so they respond.

5. Sprites Needed for the Lesson

- **Sprite 1:** Scratch Cat
- **Sprite 2:** Another character (e.g., Boy, Girl, Robot)

6. Example 1 (Step-by-Step)

Step 1: Set Up the Sprites

1. Open **Scratch 3**.
2. Add **two sprites**.
3. Place them on the Stage facing each other.

Step 2: First Sprite Speaks

1. Click **Sprite 1**.
2. Go to **Events** → drag **when green flag clicked**.
3. Go to **Looks** → add a **say** block.
4. Go back to **Events** → add **broadcast [message1]**.

when green flag clicked
say "Hello! How are you?" for 2 seconds
broadcast message1

Step 3: Second Sprite Replies

1. Click **Sprite 2**.
2. Go to **Events** → drag **when I receive [message1]**.
3. Add a **say** block.
4. Add another broadcast.

when I receive message1
say "I am fine. How about you?" for 2 seconds
broadcast message2

Step 4: First Sprite Responds Again

1. Go back to **Sprite 1**.
2. Add **when I receive [message2]**.
3. Add a **say** block.

when I receive message2
say "I am doing great too!" for 2 seconds

7. Learner Guided Practice(Example 2)

Learners should:

- Change the words spoken by the sprites.
- Add a third message.
- Change the characters to create a short story.

Example 3

Materials

- Computer with Scratch 3
- Two or more sprites (e.g., Cat and Dog)
- A backdrop (any background scene)

Step-by-Step Activity

Step 1: Set Up Sprites and Backdrop

1. Open Scratch 3 and click **Create**.
2. Add two sprites: **Cat** and **Dog**.
3. Add a backdrop of your choice (e.g., Park).

Step 2: Broadcast from Sprite 1 (Cat)

1. Select the **Cat sprite**.
2. Add this code:

```
when green flag clicked  
say "Hello! Are you ready?" for 2 seconds  
broadcast [StartConversation]
```

This sends a message called StartConversation to other sprites.

Step 3: Respond to Broadcast in Sprite 2 (Dog)

1. Select the **Dog sprite**.
2. Add this code:

```
when I receive [StartConversation]  
say "Yes! Let's begin!" for 2 seconds
```

The Dog sprite responds when it receives the broadcast.

Step 4: Extend the Conversation

- Cat can respond back after Dog:

```
when I receive [StartConversation]  
wait 2 seconds  
say "Great! Let's play!" for 2 seconds
```

- Dog can respond again if you want, creating a **conversation chain**.

8. Common Mistakes to Watch For

- Forgetting to use **broadcast** after a sprite speaks.
- Using the **same message name** incorrectly.
- Placing scripts on the **wrong sprite**.

9. Independent Practice (Short Task)

Ask learners to:

- Create a **conversation with at least 4 lines**.
- Use **at least 2 broadcasts**.
- Make sure sprites speak in the correct order.

10. Assessment Questions

1. What is a broadcast in Scratch?
2. Which block receives a broadcast?
3. Why are broadcasts important in conversations?
4. What happens if no sprite receives a broadcast?

11. Summary

- Broadcasts help sprites **communicate**.
- Conversations need **messages** to control order.
- Broadcasts make Scratch projects **interactive**.
- Conversations can be used in **stories, games, and animations**.

12. Extension (Fast Learners)

- Add **sound effects** to each speech.
- Change costumes when a sprite speaks.
- Create a **conversation with 3 sprites**.

Assignment Project: Here is a guiding question for the students' assignment project:

- Imagine you are going back to school, from your one month vacation with your 2 classmates who are your friends. Assuming they went for the vacation in different countries and you went to a different one as well. Create a conversation that you would have amongst yourselves while moving to school (use coordinates and Broadcasts).